

Figure 1 displays the fraction of simulated supernovae that are detected (left panel) and the median detection radius $\mathcal{R}_m$ (right panel) for various supernova models. The left panel is a horizontal bar chart where the x-axis represents the fraction of detected supernovae, ranging from 0.0 to 1.0. The y-axis lists 100 different supernova models, each identified by a name, code, and parameters. The bars are colored blue for fiducial models and orange for non-fiducial models. The right panel is a scatter plot showing the median detection radius $\mathcal{R}_m$ in $\text{Gpc}^{-3} \text{yr}^{-1}$ on a logarithmic scale from 10^1 to 10^2 . The points are colored blue for fiducial models and orange for non-fiducial models. A grey shaded region in the right panel indicates the detection radius range for the fiducial model.